I

(1)

The time-independent Schrödinger equation (TISE) is

$$-\frac{\hbar^2}{2m}\frac{d^2\phi(x)}{dx^2}+V(x)\phi(x)=E\phi(x).$$

(2)

Since $V(x)\to\infty$ when $|x|>\frac{a}{2}$, the wavefunction vanishes outside the potential well:

$$\phi(x)=0,\qquad |x|>\frac{a}{2}.$$

Considering the continuity of the wavefunction, we obtain the boundary conditions

$$\boxed{\phi(a/2)=\phi(-a/2)=0}.$$

(3)

Within the potential well, where $V(x)=0$, the TISE becomes

$$-\frac{\hbar^2}{2m}\frac{d^2\phi(x)}{dx^2}=E\phi(x).$$

Let

$$k=\frac{\sqrt{2mE}}{\hbar},$$

then the equation can be written as

$$\frac{d^2\phi(x)}{dx^2}+k^2\phi(x)=0.$$

Its general solution is

$$\phi(x)=A\cos(kx)+B\sin(kx).$$

Noticing that $V(x)=V(-x)$, we have

$$[P,H]=0,$$

which means that the eigenstates of the Hamiltonian can be chosen to have definite parity.

For even-parity solutions,

$$\phi(x)=A\cos(kx).$$

Considering the boundary conditions obtained in (2), we have

$$\cos\left(\frac{ka}{2}\right)=0,$$

which gives

$$k = n_{\text{odd}} \frac{\pi}{a},$$

where n_{odd} denotes a positive odd integer.

Similarly, for odd-parity solutions,

$$\phi(x) = B \sin(kx),$$

and the boundary conditions give

$$\sin\left(\frac{ka}{2}\right) = 0,$$

which leads to

$$k = n_{\text{even}} \frac{\pi}{a},$$

where n_{even} denotes a positive even integer.

For the bounded states, we impose the normalization condition

$$\int_{-a/2}^{a/2} \phi^*(x)\phi(x)\, dx = 1.$$

Therefore,

$$\boxed{A = B = \sqrt{\frac{2}{a}}}.$$

The normalized eigenfunctions can thus be written as

$$\phi_n(x) = \begin{cases} \sqrt{\dfrac{2}{a}} \cos\left(\dfrac{n\pi x}{a}\right), & n \text{ odd}, \\ \sqrt{\dfrac{2}{a}} \sin\left(\dfrac{n\pi x}{a}\right), & n \text{ even}. \end{cases}$$

The energy eigenvalues are

$$\boxed{E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}}, \qquad n = 1, 2, 3, \ldots$$

(4)

Expand the initial wavefunction in terms of the energy eigenfunctions of the Hamiltonian:

$$\psi_0(x) = \sum_{n=1}^{\infty} c_n \phi_n(x).$$

Since

$$\psi_0(x) = \psi_0(-x),$$

the initial wavefunction has even parity. Therefore, its expansion coefficients vanish for all odd-parity eigenfunctions.

In our notation, the even-parity eigenfunctions correspond to odd values of n, while the odd-parity eigenfunctions correspond to even values of n.

Using

$$c_n = \langle \phi_n | \psi_0 \rangle,$$

in the position representation, we have, for odd n,

$$c_n = \sqrt{\frac{2}{a}} \int_{-a/2}^{a/2} \cos\left(\frac{n\pi x}{a}\right) \psi_0(x)\, dx.$$

Substituting the given initial wavefunction and evaluating the integral, we obtain

$$\boxed{c_n = \frac{8\sqrt{15}}{n^3\pi^3}(-1)^{\frac{n-1}{2}}}, \qquad n \text{ odd}.$$

For even n, $c_n = 0$.

Thus, the probability of measuring the energy eigenvalue E_n is

$$\boxed{|c_n|^2 = \begin{cases} \dfrac{960}{n^6\pi^6}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}}$$

(5)

Using the probabilities obtained in (4), the expectation value of the energy is

$$\begin{aligned} \langle H \rangle &= \sum_{j=1}^{\infty} |c_{2j-1}|^2 E_{2j-1} \\ &= \frac{480\hbar^2}{ma^2\pi^4} \sum_{j=1}^{\infty} \frac{1}{(2j-1)^4} \\ &= \frac{480\hbar^2}{ma^2\pi^4} \cdot \frac{\pi^4}{96}. \end{aligned}$$

Therefore,

$$\boxed{\langle H \rangle = \frac{5\hbar^2}{ma^2}}.$$

Similarly, the expectation value of the squared energy is

$$\begin{aligned} \langle H^2 \rangle &= \sum_{j=1}^{\infty} |c_{2j-1}|^2 E_{2j-1}^2 \\ &= \frac{240\hbar^4}{m^2a^4\pi^2} \sum_{j=1}^{\infty} \frac{1}{(2j-1)^2} \\ &= \frac{240\hbar^4}{m^2a^4\pi^2} \cdot \frac{\pi^2}{8}. \end{aligned}$$

Thus,

$$\boxed{\langle H^2 \rangle = \frac{30\hbar^4}{m^2 a^4}}.$$

The standard deviation of the energy is

$$\begin{aligned}\Delta E &= \sqrt{\langle H^2 \rangle - \langle H \rangle^2} \\ &= \sqrt{\frac{30\hbar^4}{m^2 a^4} - \frac{25\hbar^4}{m^2 a^4}}.\end{aligned}$$

Therefore,

$$\boxed{\Delta E = \frac{\sqrt{5}\hbar^2}{m a^2}}.$$

(6)

The mistake is that the differential expression of the Hamiltonian is applied twice only within the potential well, without properly considering the boundary conditions.

Although $\psi_0(x)$ vanishes at the boundaries, applying the Hamiltonian once gives

$$H\psi_0(x) = \frac{\hbar^2}{m}\sqrt{\frac{30}{a^5}},$$

which is a nonzero constant within the well.

This function does not satisfy the boundary conditions of the infinite potential well. Therefore, the Hamiltonian cannot simply be applied to it again as an ordinary second derivative within the well.

The correct second moment can instead be obtained from the energy eigenstate expansion:

$$\boxed{\langle H^2 \rangle = \sum_{n=1}^{\infty} |c_n|^2 E_n^2}.$$

(7)

The time-dependent Schrödinger equation (TDSE) is

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = H\psi(x,t).$$

By applying the method of separation of variables to each energy eigenstate, we obtain

$$\psi_n(x,t) = \phi_n(x) T_n(t).$$

The time-dependent factor satisfies

$$T_n(t) = e^{-iE_n t/\hbar}.$$

Therefore, using the expansion coefficients obtained in (4), the wavefunction satisfying the initial condition is

$$\boxed{\begin{aligned}\psi(x,t) = \sum_{j=1}^{\infty} \frac{8\sqrt{15}}{(2j-1)^3\pi^3}(-1)^{j-1}\sqrt{\frac{2}{a}} \\ \times \cos\left(\frac{(2j-1)\pi x}{a}\right) \exp\left[-i\frac{(2j-1)^2\pi^2\hbar}{2ma^2}t\right].\end{aligned}}$$

This expression applies within the potential well, $-a/2 \leq x \leq a/2$, and the wavefunction vanishes outside the well.

II

(8)

For $0 \leq x \leq a$, the TISE is

$$-\frac{\hbar^2}{2m}\frac{d^2\phi(x)}{dx^2} + V_0\phi(x) = E\phi(x).$$

Let

$$k_2 = \frac{\sqrt{2m(E-V_0)}}{\hbar}.$$

The general solution is

$$\phi(x) = C\cos(k_2 x) + D\sin(k_2 x).$$

Considering the boundary condition at $x = 0$,

$$\phi(0) = 0,$$

we obtain $C = 0$.

Thus,

$$\boxed{\phi(x) = D\sin(k_2 x)}.$$

(9)

At $x = a$, we impose the continuity conditions for the wavefunction and its first derivative.

Using the wavefunction in the region $x > a$,

$$\phi(x) = B\left(e^{-ik_3 x} + e^{2i\delta}e^{ik_3 x}\right),$$

we obtain

$$\begin{cases} D\sin(k_2 a) = B\left(e^{-ik_3 a} + e^{2i\delta}e^{ik_3 a}\right), \\ Dk_2\cos(k_2 a) = ik_3 B\left(-e^{-ik_3 a} + e^{2i\delta}e^{ik_3 a}\right). \end{cases}$$

Dividing the second equation by the first, we obtain

$$k_2 \cot(k_2 a) = ik_3 \frac{-e^{-ik_3 a} + e^{2i\delta} e^{ik_3 a}}{e^{-ik_3 a} + e^{2i\delta} e^{ik_3 a}}.$$

Using the identity

$$i\frac{e^{2i\theta} - 1}{e^{2i\theta} + 1} = -\tan\theta,$$

with $\theta = \delta + k_3 a$, we get

$$k_2 \cot(k_2 a) = -k_3 \tan(\delta + k_3 a).$$

Therefore,

$$\tan(\delta + k_3 a) = -\frac{k_2}{k_3} \frac{1}{\tan(k_2 a)}.$$

Finally,

$$\boxed{\delta = -k_3 a - \arctan\left(\frac{k_2}{k_3} \frac{1}{\tan(k_2 a)}\right) \pmod{\pi}}.$$

(10)

The right-hand side represents a superposition of an incident wave and a reflected wave.

The reflection amplitude is

$$r = e^{2i\delta}.$$

Therefore, the reflectivity is

$$R = |r|^2 = 1.$$

This means that the particle is perfectly reflected, and the reflected wave acquires a phase shift of 2δ relative to the incident wave.